

NIKUNJ AGRAWAL

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Education

University of Pennsylvania

Philadelphia, PA, United States

Master of Computer and Information Technology

(Aug 2024 – May 2026)

Relevant Coursework: Machine Learning, Software Systems (Distributed Systems), Databases, Cybersecurity, Big Data Analytics, DSA

Research Assistant: Data Researcher at the Center for Engineering Mechanobiology (Funded by NSF)

Shri Ram College of Commerce (SRCC), University of Delhi / GPA: 8.68/10 (Top 5%)

New Delhi, India

Bachelor of Commerce (Honors) (Major in Finance & Economics)

(July 2018 – May 2021)

Honors: V.K. Bhalla Gold Medal in Financial Management, CMA Scholarship for the full program

Activities: President (National Service Scheme), Secretary (Student Government of Class of 2021), Computer and Maths Society

Professional Experience

Founder's Office – Product & Strategy / Darwinbox (HR SaaS Unicorn) – Hyderabad, India

Dec 2021 – Jan 2023

- **Led technical product management** for US market expansion, owning end-to-end lifecycle (discovery → PRD → MVP → production) for agentic AI and integration-heavy features that increased **user adoption by 25%**
- Translated customer & payroll-partner requirements into **product specs** and API-level feature requirements
- **Architected third-party API integrations** with 3 major US payroll providers, defining data models, error taxonomies, and rollout plans to expand platform capabilities and enterprise readiness
- Conducted market and **competitor analysis with EY's HR consulting team**, informing roadmap decisions that drove a **20% increase in market share** outside India/SEA
- Supported in a **\$72M Series-D Unicorn** funding round by contributing to product narratives, metrics, and **technical diligence**
- Owned production **RAG-based AI agent**—defining prompt patterns, tool-use flows, **evaluation criteria**, and **output guardrails**—integrated with React.js to accelerate knowledge retrieval and boost user productivity

Associate Consultant / GDi Partners (Tech & Government Consulting) – Delhi, India

Jan 2023 - Feb 2024

- **Product** Consultant for a **5+ Year & \$400M+** Agriculture project for a state funded by the **Bill and Melinda Gates Foundation**
- **Owned end-to-end delivery** of a Python-based AI workflow platform for issue routing, automated triage, and human-in-the-loop grievance redressal across government education programs
- Defined system workflows, data requirements, success metrics, and evaluation cadence, achieving a **30% increase in user engagement** and **80% automated issue resolution within 48 hours**
- Leveraged **predictive data models** to improve learning outcomes for **Grades 5–8 by 28%**, and **15% reduction in dropouts**
- Acted as the bridge between policymakers, domain experts, and technical teams to translate ambiguous requirements into solutions

ML and Data Intern / Provost's Office, University of Pennsylvania – Philadelphia, PA

May 2025 - Aug 2025

- Partnered with academic stakeholders to define **data requirements and success metrics** for analyzing enrollment patterns across **10%+ of UPenn's course catalog**
- Delivered predictive insights and **dashboards** in **Tableau** to support data-driven decision-making for faculty and administrators
- Developed a customer-facing **REST API** for **multi-platform course lists** with real-time tracking and comprehensive error handling

Technical Product Analyst / Astra Asset Management UK Ltd. – London & Delhi

June 2021 – Dec 2021

- Supported **Python** and **MERN-based financial models** to support investment decisions and optimize portfolio performance
- Enhanced scalability and performance by containerizing financial solutions with **Docker** and deploying on **Microsoft Azure**
- Led a **Bayesian optimization-based multi-objective calibration pipeline** using BoTorch with Gaussian Process surrogates

Projects

Predictive Risk Modeling System (Meteorite Fall Prediction)

- Defined the problem scope and **success metrics** for predicting meteorite fall probability using historical data for an **ML pipeline**
- Evaluated multiple modeling approaches (**Logistic Regression, Random Forest, Gradient Boosting, Ensembles**) to balance **interpretability, accuracy**, and **operational feasibility** using **precision, recall, F1-score, and ROC-AUC**
- Addressed data imbalance using **BorderlineSMOTE**, improving minority class recall for rare-event prediction use cases
- Synthesized model outputs into **stakeholder-ready insights** explaining **risk, uncertainty, and confidence levels**

RAG-Based Financial News Intelligence Platform

- Used RAG to conceptualize and build a product to **answer natural-language queries** on over **10+ years** of financial news data on the S&P 500 stocks using embedding-based retrieval tools like **DistilBERT**
- Made tradeoff decisions between keyword search, **embedding-based retrieval**, and response latency to optimize user experience
- Designed explainable outputs linking **historical events, source citations**, and **price movement context**
- Integrated contextual retrieval and extended the system to analyze price movements and correlations across related equities

Leadership, Competitions & Research

- **Co-President** of the Pan Asian Graduate Students Assembly (PAGSA) at UPenn, having an **annual budget of \$50K+**
- **Vice President** of the National Service Scheme - the largest student-run club on campus during Undergraduate studies
- Recognized as **Best Social Worker of the Year** for commitment to **Community Service** and **Social Entrepreneurship**
- **Awarded 1st Prize** in the National Case Study Competition at **the Indian Institute of Technology (IIT), Roorkee**.

Technical Skills

- **Product & Analytics:** Product roadmapping, PRDs, metrics definition, AI evaluation frameworks, human-in-the-loop design,
- **Programming & Data:** Python, C, C++, SQL, Java, scikit-learn, TensorFlow, Pandas, NumPy, HTML/CSS
- **Systems and Tools:** LangChain, RAG pipelines, LLM observability, multi-agent orchestration, APIs, backend services, databases (SQL, MongoDB, Vector DBs), ML pipelines, Git, Docker, Tableau, AWS. Agentic workflow design